

State Transition Testing

A technique for validating behavior when outcomes depend on the current state and occurring events.

Model

Identify states, events, valid/invalid transitions and resulting actions.

Examples

Order: created → paid → shipped → delivered; account: active → locked → reactivated; session: authenticated → expired.

Coverage

Test valid paths, forbidden events, repetition, wrong order and final states.

History

Defects may depend on previous sequence; record initial state and events for reproducibility.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

QA tip

When the same event produces different outcomes depending on current state, model transitions explicitly.